

Social Media Misinformation Detection

Module: Propagation and Influence (Graph)

Literature Review

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1 INTRODUCTION AND SCOPE

Misinformation is defined as much by how it travels as by what it says. This review addresses the network layer of the project: how a post propagates across the social graph, how propagation structure helps separate misleading content from reliable content, and which accounts are most responsible for the spread. It is one of two modules; the companion module classifies the credibility of a post from its text. The two are designed to operate independently. This module runs on the veracity labels its datasets already provide, and attaching the companion module's per-post credibility to the graph nodes is a final integration step rather than a prerequisite.

The review proceeds from the empirical case for modelling propagation, through the datasets that record social context, the modelling of diffusion from classical cascades to graph neural networks, the temporal dynamics of cascades, the scoring of account influence and its connection to intervention, and the visualisation of spread. A closing synthesis states the choices adopted for this module.

The analysis is conducted in English. The project additionally incorporates a small sample of real, collected data, used mainly to validate the influence ranking on observed cascades.

2 PROBLEM FRAMING AND MOTIVATION

A few terms recur. A *cascade* is the tree or graph of reshares produced as a post spreads; a *rumour* is an unverified claim in circulation; *propagation* and *diffusion* denote the process by which content moves between accounts (Zubiaga et al., 2018). The network module operates on these objects rather than on text.

The empirical case for studying propagation is strong. Vosoughi et al. (2018) analysed roughly 126,000 rumour cascades on Twitter spread by about three million people and found that false news travelled farther, faster, and more broadly than the truth, that falsehoods were about 70% more likely to be reshared, and that the effect was driven by people rather than by automated accounts. The way a claim spreads therefore carries signal about whether it is true, and that signal is available even when the text is short, ambiguous, or absent. This is the premise on which propagation-based detection rests.

3 DATASETS WITH SOCIAL CONTEXT

This module requires more than text: it requires who posted, who reshared, and the structure of the resulting cascade. Table 1 lists the principal resources that record this context.

Dataset	Social context	Labels	Best fit
FakeNewsNet	User engagement, relations	Fake / real	Content + propagation
PHEME	Conversation threads	Rumour / non-rumour	Rumour structure
Twitter15/16	Propagation trees	4-way veracity	GNN benchmarks
CREDBANK	~60M tweets, 1,000+ events	5-point credibility	Event-level diffusion
UPFD (PyG)	Propagation graphs + user history	Fake / real	Ready-made GNN input

Table 1: Datasets that record the social context needed for propagation modelling.

FakeNewsNet records news items alongside user engagement and relations, which makes it well suited to a project that works at both the content and network levels (Shu et al., 2020). PHEME captures rumours and non-rumours as conversation threads from breaking-news events, preserving the reply structure (Zubiaga et al., 2016). The Twitter15 and Twitter16 collections, propagation trees with four-way veracity labels, are the standard benchmarks for graph-based rumour detection and are the sets on which Bi-GCN and comparable models are evaluated (Bian et al., 2020). CREDBANK is a large credibility-annotated corpus suited to event-level diffusion (Mittra and Gilbert, 2015). The UPFD release is of particular practical value: it provides propagation graphs with user-history features already prepared and integrated into PyTorch Geometric (Dou et al., 2021; Fey and Lenssen, 2019), removing much of the data-engineering cost of an initial model.

A practical constraint follows from this table: propagation methods depend on social-context data, which is scarcer and harder to collect than text, and access to it has tightened as platform APIs have closed. This is a further reason to build on datasets that already package the propagation structure.

4 GRAPH CONSTRUCTION AND NODE FEATURES

Before any model is trained, the raw social data must be turned into a graph. The usual choice is to represent posts and users as nodes and to draw edges from reshares, replies, and follow relations, yielding either a per-story propagation tree or a larger heterogeneous graph that mixes user and content nodes. Node features combine several sources: a text representation of the post, which in this project can come from the content module’s encoder; user attributes such as account age, activity level, and follower count; and timestamps recording when each interaction occurred.

PyTorch Geometric expresses each cascade as a graph object with node features, an edge list, and a label, and supports mini-batching over many such graphs (Fey and Lenssen, 2019). The UPFD release follows this format, pairing propagation graphs with user-history features, which makes it a convenient template for the project’s own data (Dou et al., 2021). The main engineering effort lies in reconstructing cascades from raw logs and in keeping the feature set consistent between public datasets and any collected sample.

5 MODELLING PROPAGATION

Modelling how information spreads predates deep learning. The independent cascade and linear threshold models describe diffusion as a stochastic process over a graph, in which an informed node activates its neighbours with some probability. Kempe et al. (2003) formalised influence

maximisation on top of these models, selecting a seed set to maximise expected spread, and showed that a greedy algorithm attains a constant-factor guarantee for this submodular objective. These models, related to the epidemic models of contagion, supply a vocabulary for reasoning about cascades, seeds, and reach, even though the task here is detection and ranking rather than seeding.

The first learning-based methods treated a story’s spread as a sequence. Ma et al. (2016) applied a recurrent network to the time series of a rumour’s posts, and Ma et al. (2018) moved to tree-structured recursive networks that follow the propagation tree directly, capturing structure that a flat sequence omits. This progression, from text alone, to the time series, to the tree, leads to the graph models of the next section.

6 GRAPH NEURAL NETWORKS FOR DETECTION

Graph neural networks generalise the tree models to arbitrary propagation graphs. GCN defines a convolution over graph neighbourhoods (Kipf and Welling, 2017), and GAT replaces fixed neighbour weights with learned attention (Veličković et al., 2018); together they are the standard building blocks. Bi-GCN is the representative misinformation model: it reads the cascade both top-down, as propagation, and bottom-up, as dispersion, and reports accuracy around 0.886 on Twitter15/16 (Bian et al., 2020). UPFD adds a user-preference signal, fusing the engaged users’ history with the propagation graph through a GNN encoder (Dou et al., 2021). GCAN is a related model that also produces explanations, highlighting suspicious resharers and the terms they emphasise (Lu and Li, 2020). Reference implementations of several of these detectors are collected in the safe-graph GNN-FakeNews repository (safe-graph, 2021), and PyTorch Geometric is the library on which this module builds (Fey and Lenssen, 2019).

A natural progression is to begin from a GCN/GAT baseline, advance to a Bi-GCN-style bidirectional model, and, because UPFD ships with prepared data in PyTorch Geometric, adopt it as a low-friction way to introduce user-level signal. It is also where the content module can plug in later: its credibility score can weight or relabel graph nodes, although the detector trains on the datasets’ own labels without it.

It helps to situate these models on a spectrum. Purely content-based detectors read only the text and ignore who spreads it; purely structural detectors read only the cascade and ignore what is said; hybrid models such as UPFD and GCAN combine the two, which is generally the stronger setting and the one this project adopts across its two modules (Dou et al., 2021; Lu and Li, 2020). Within the structural family, the practical differences lie in the direction of message passing, top-down, bottom-up, or both, and in whether user features beyond the graph are included, which is where the choice between a plain GCN, a Bi-GCN, and UPFD effectively sits.

7 TEMPORAL DYNAMICS AND EARLY DETECTION

Propagation is not static, and timing carries signal. The cascades of false stories not only reach more people but reach them faster, traversing the network in less time than true ones (Vosoughi et al., 2018). The early sequence models captured this implicitly by reading the post time series (Ma et al., 2016), and the structural models inherit it through the shape of the cascade as it grows.

A recurring practical question is how early a reliable verdict can be reached. A detector that is accurate only after a story has fully spread offers little operational value, so accuracy as a function of elapsed time, or of the number of posts observed, is a meaningful axis alongside final accuracy. The temporal profile of a cascade, its speed and depth over time, is therefore treated both as a feature for the model and as a constraint the evaluation should reflect.

Two framings of time recur and are complementary. In the first, time is an input: inter-post intervals, cascade depth, and growth rate become features the model reads. In the second, time is an evaluation condition: the model is restricted to the opening hours of a cascade and scored on how well it decides under that limit. A system intended for practical use should be reported under both.

8 ACCOUNT INFLUENCE, SUPERSPREADERS, AND INTERVENTION

The third question is which accounts drive the spread. Two strands of work inform it. The structural strand ranks nodes by network centrality, degree, betweenness, PageRank, and k-core position; Kitsak et al. (2010) found that a node’s position in the k-core decomposition is a better guide to its spreading power than its number of connections. The empirical strand studies observed diffusion directly: Pei et al. (2014) located superspreaders from real cascades, and later work on low-credibility content found that a small number of accounts drive a large share of the spread and that structural centrality alone is insufficient, with behavioural traits also mattering (DeVerna et al., 2024). The role of automated accounts is part of this picture, though the large-scale evidence indicates that human resharing, not bots, accounts for most of the differential spread of false content (Vosoughi et al., 2018).

Influence scoring connects naturally to intervention. Limiting the spread of misinformation can be framed as the inverse of influence maximisation: where Kempe et al. (2003) selects seeds to maximise reach, containment selects accounts whose removal or correction most reduces it, and the k-core and centrality rankings above identify candidate targets. A practical scheme combines inexpensive structural measures, k-core and PageRank, with simple behavioural features such as posting volume and reshare rate, then validates the resulting ranking against observed cascades. What this module owes the project is a ranked shortlist of the accounts that move misinformation, which is more actionable than any single global score.

Spread is shaped as well by the structure of the network around a source. Misinformation often circulates within densely connected communities whose members share outlooks, so a claim can travel far inside a cluster while barely crossing into others. Accounts that bridge otherwise separate communities are thus disproportionately important to spread, a role that bridging-oriented measures such as betweenness capture better than local popularity, and one a ranking should not overlook.

9 VISUALISATION

A propagation analysis is only useful if its results can be read. Hoaxy is a reference point: it collects and visualises how claims and their fact-checks spread on Twitter, rendering the diffusion network in which nodes are accounts and edges are reshares (Shao et al., 2016). For offline analysis, Gephi is the standard open-source tool for laying out and inspecting large networks (Bastian et al., 2009). For this project, a focused web dashboard that displays a cascade, highlights the top-ranked spreaders, and overlays the content module’s credibility scores is a realistic deliverable, with Gephi serving the exploratory analysis that precedes it.

10 LIMITATIONS AND OPEN CHALLENGES

Propagation-based methods face several recognised constraints. They depend on social-context data, which is harder to obtain than text and has become less accessible as platform interfaces have restricted automated collection. A model trained on the cascades of one set of events, or one platform, may transfer poorly to another, since propagation patterns are shaped by platform mechanics and audience. Labels remain scarce, because annotating veracity at the level of a story

is costly, and benchmark cascades are often small, which limits how much structure a model can exploit. There is also a tension between accuracy and timeliness: the most informative propagation features accumulate as a story spreads, yet detection is most valuable early, before a cascade is large. These constraints frame the choices below and set realistic expectations for the module.

11 SCALABILITY

Real propagation graphs can be large, and the engineering must account for it. Per-story cascades are individually small and batch well, but a global graph of users and their relations can reach millions of nodes, beyond what fits on a single device for full-graph training. The standard response is neighbour sampling, which trains on sampled subgraphs around each target node rather than on the whole graph, and PyTorch Geometric provides loaders for both the per-cascade and the sampled-subgraph regimes (Fey and Lenssen, 2019). For this project the per-cascade setting is the starting point, since it matches the benchmark datasets and keeps memory modest, with sampled training held in reserve should a larger graph be assembled.

12 SYNTHESIS: CHOICES FOR THIS MODULE

The literature supports a coherent plan, summarised in Table 2.

Task	Approach	Data / metric
Detection	GCN/GAT baseline → Bi-GCN; consider UPFD	FakeNewsNet, UPFD (PyG); accuracy, macro-F1
Early detection	Accuracy vs. elapsed time / posts observed	Cascade time profile
Influence scoring	k-core + PageRank with behavioural features	Datasets with cascades; top- <i>k</i> spreaders, rank correlation
Visualisation	Web dashboard of cascades + top spreaders; Gephi for analysis	Project data; qualitative

Table 2: Recommended approach for the propagation-and-influence module.

In summary, the module builds propagation graphs in PyTorch Geometric, advancing from a GCN/GAT baseline to a Bi-GCN-style model and optionally to UPFD. Account influence is scored by combining k-core and PageRank with behavioural features and validated against observed cascades, and the results are presented through a dashboard that can overlay the content module’s credibility scores on the network. Detection is assessed by accuracy and macro-F1, with early detection tracked as a function of time. The two modules remain conceptually joined, content characterises what is misleading and the network characterises how it travels and who carries it, while each remains independently runnable.

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